

HA NGUYEN, Ph.D.

360-771-2424 • nvha.dvt@gmail.com • nvha.github.io • linkedin.com/in/nvha • Phoenix, AZ (US person – Green card)

SUMMARY

Hardware Systems Architect and Senior Network Systems Architect (Ph.D.) with 3+ years of dedicated post-doctoral R&D alongside 5+ years of university research leadership bridging high-throughput FPGA hardware acceleration with scale-out network protocol design. On the hardware side, brings deep expertise in full-stack RTL design (SystemVerilog/Verilog), high-speed DSP pipelines, ultra-wide 1024-bit/512-bit AXI-Stream datapaths, PCIe 3.0 DMA offload engines, and multi-vendor FPGA prototyping across Intel Agilex and AMD Xilinx architectures. On the networking side, specializes in high-speed transport protocol optimization (TCP/NC, RoCE/RDMA), Software-Defined Networking (SDN/OpenFlow), P4 in-band telemetry, active queue management (AQM), and discrete-event simulation in ns-3 and Mininet. Backed by 40+ peer-reviewed publications (IEEE/IEICE) and over \$100,000 in R&D grants, excels at hardware-software co-design to translate complex network control logic into low-latency physical silicon and programmable data planes.

TECHNICAL SKILLS

Core Domains : FPGA & ASIC Design, RTL Design, Hardware Emulation, Hardware Verification, Pre/Post-Silicon Validation, HW/SW Co-Design, DSP Algorithms, Packet Processing, FEC Architectures, High-Speed Datapaths, Network Modeling & Simulation, Transport Protocols (TCP/IP, UDP, TCP/NC, RoCE/RDMA), Software-Defined Networking (SDN / OpenFlow), Flow & Congestion Control, Active Queue Management (AQM), Packet Loss Masking, Tail-Latency Reduction, Adaptive Routing & Load Balancing, Programmable Data Planes (P4), Traffic Engineering

Software & Scripting : SystemVerilog, Verilog, P4, C/C++, Python, TCL, MATLAB, Simulink, Bash, Git, Linux (Ubuntu, PetaLinux, Embedded), Agile

Hardware & Protocols : FPGAs (Intel Agilex/Stratix, AMD Xilinx Kintex UltraScale+), TI LMK62E2 Oscillator, Microcontrollers, Synopsys Symphony, Altera DSP Builder, PCIe 3.0, DMA, SerDes, Ethernet MAC/PHY, AXI-Stream, APB, Digital Logic Synthesis

Systems & Simulation : Vivado (Block Designer), Quartus (Platform Designer), ModelSim, ns-3 Network Simulator, Mininet, RYU OpenFlow Controller, BMv2 Software Switch, GNS3, Packet Tracer, Wireshark

EXPERIENCE

University of Science

Oct. 2021 – Apr. 2026

Network Systems Researcher & Laboratory Supervisor

Ho Chi Minh City, Vietnam

- Directed major R&D initiatives within the ICTLAB focusing on high-performance networking, wireless communications, and industrial sensors, architecting loss-masking transport frameworks, proactive FEC queue management models, and fault-tolerant TDMA scheduling.
- Led R&D projects on P4 data-plane telemetry, active queue management (AQM), adaptive SDN routing optimization/load balancing, and IoT network protocol modeling.
- Co-founded the ICTLAB, bridging theoretical transport protocol design, wireless network modeling, and DSP algorithms with FPGA prototyping and C++/Python discrete-event network modeling in ns-3.
- Secured and managed over \$100,000 in competitive industry and government R&D funding across 7 projects (including VNU-HCM Class-B Grant B2023-18-01 and an Industry R&D Collaboration), authoring 20+ international peer-reviewed papers (IEEE/IEICE/Scopus).

Kyushu Institute of Technology

Apr. 2018 – Sep. 2021

Principal Postdoctoral Researcher & PI

Fukuoka, Japan

- Served as Principal Investigator (PI) for two prestigious JSPS KAKEN Grants-in-Aid for IoT reliability and "self-healing" network projects (Grant IDs: 18H06467 & 20K19785, totaling 4.42M JPY), leading 3 years of dedicated post-doctoral protocol R&D.
- Headed R&D projects to engineer resilient TCP with Network Coding (TCP/NC) transport frameworks and centralized SDN architectures tailored to actively mask lossy transport channels, prevent TCP retransmission timeouts, and reduce tail latency.
- Developed high-level software models and discrete-event simulations in C++ and Python to bridge theoretical network capacity algorithms with practical, real-time packet processing, queue management, and hardware-applicable architectures.
- Designed and delivered specialized guest lectures on SDN architectures, OpenFlow control mechanisms, Ryu controllers, and Mininet programmable network topologies for the City College of New York (CCNY).

- Architected resilient end-to-end protocols and SDN control logic, performing real-time traffic engineering and performance characterization for high-density network deployments.

University of Science

Associate Lecturer & Network Administrator

Sep. 2009 – Mar. 2014
Ho Chi Minh City, Vietnam

- Conducted applied physical-layer research, architecting an FPGA-based 2×2 MIMO-OFDM SDM Transceiver achieving 400 Mbps throughput using custom DSP pipelines to test hardware-level forwarding performance.
- Managed departmental network infrastructure as Administrator, administering L2/L3 routing (OSPF/BGP), switching, VLAN segmentation, and network security configurations in live production environments.
- Maintained active certifications as a CCAI (Cisco Certified Academy Instructor) for CCNA/Security, mentoring student teams to a 2nd-place national finish in the Cisco NetRiders contest (2013).

MASTER PROJECT REPOSITORY

High-Speed Hardware-in-the-Loop (HITL) Link Emulation

Apr. 2026 – Present

- **Tools:** Intel Agilex 5 (DE25), Xilinx Kintex UltraScale+ (KU5P), TI LMK62E2, SystemVerilog, PCIe 3.0, DMA, HSMC, SMA, Ethernet.
- Architected a heterogeneous hardware-in-the-loop (HITL) emulation system bridging Intel Agilex 5 and Xilinx Kintex UltraScale+ FPGAs over multi-vendor interfaces (HSMC-XTS, SFP-to-SMA, PCIe 3.0x4).
- Utilized a TI LMK62E2 high-performance oscillator driving an industry-standard 156.25 MHz SerDes reference clock to lock transceivers and synchronize physical layer links.
- Designed an ultra-wide 1024-bit packet processing datapath operating on an industry-standard 156.25 MHz / 312.5 MHz core clock (up to 320 Gbps internal pipeline capacity), executing real-time header parsing, deparsing, field modification, and packet filtering.
- Engineered host-to-FPGA streaming pipelines via PCIe 3.0x4 DMA, enabling end-to-end packet processing while architecting embedded Network Coding (NC) encoder/decoder offload logic.

Wireless System Emulation: AWGN Hardware Emulator

Sep. 2025 – Feb. 2026

- **Tools:** Quartus, Vivado, SystemVerilog, Modelsim, TCL, Python, C, Git.
- Architected a resource-optimized, real-time embedded AWGN hardware generator supporting multi-rates and multi-users, integrated with an APB interface for dynamic C/C++ SDN software control.
- Designed 512-bit wide AXI-Stream datapaths achieving 375MHz+ (up to 400MHz max operating frequency) on Intel Agilex 7 and Xilinx FPGAs, optimizing strict latency requirements for physical silicon translation.
- Developed scalable control state machines, low-level drivers, and high-level software to control high-fidelity noise generation, sweeping SNR from –30 dB to 45 dB with precise real-time configuration.
- Automated build flows by authoring custom TCL scripts for project generation and pin assignments, streamlining model usability and continuous pre-silicon integration.
- Facilitated HW/SW co-design by implementing APB master/slave interfaces via Vivado Block Designer and Intel Platform Designer, providing memory-mapped register access for software validation teams via C drivers.

Programmable Data Plane (P4) for Monitoring & Loss Mitigation

Jan. 2024 – Jan. 2025

- **Tools:** P4, BMv2 Switch, ns-3, C++, Python, Mininet.
- Architected and validated a hop-by-hop packet loss recovery scheme for reliable multicast using P4-enabled switches, achieving a 36% reduction in transfer times under severe link loss via ns-3 simulations.
- Engineered a custom 8-byte Control Header featuring a 40-bit Success_Check sequence to provide in-band telemetry, tracking packet arrival jitter and queue occupancy with microsecond precision.
- Implemented data-plane packet buffering via a P4 recirculation port queue, enabling intermediate switches to trigger localized retransmissions without central controller intervention.
- Programmed foundational P4 pipelines on the BMv2 software switch to dynamically manage network flow control and queue depths.

Fault-Tolerant TDMA Framework for LWSNs & Constellation Networks

Oct. 2023 – Sep. 2025

- **Tools:** ns-3, C++, Python.
- Performed protocol-level algorithm development to architect a proactive packet loss recovery framework, integrating hop-by-hop Forward Error Correction (FEC) / Erasure Coding with TDMA scheduling.
- Utilized a 1-step Hill Climbing search algorithm to optimize TDMA timeslots and maintain >99% reliability under aggressive 50% link-loss environments.
- Programmed scalable software models in C++ and Python to simulate data paths over a 20-mile multi-hop linear topology to optimize timeslot allocations.

Self-Healing Dynamic Routing Protocol (SDN-FEDRP)

Jan. 2021 – Dec. 2021

- **Tools:** OpenFlow, RYU Controller, Mininet, Python.

- Architected a dynamic routing protocol for SDN to actively optimize load balancing, manage real-time link utilization, and prevent data flow congestion across complex topologies.
- Engineered a custom routing algorithm on the Python-based RYU controller leveraging Depth First Search (DFS) to discover paths and calculate costs dynamically.
- Implemented an automated OpenFlow reallocation mechanism governed by a fairness factor, validating via Mininet and achieving a 7x improvement in load balancing versus standard SDN-OSPF.

TCP with Network Coding (TCP/NC)

Mar. 2017 – Mar. 2020

- **Tools:** ns-3, C++, Python.
- Engineered resilient embedded transport layer state machines and a custom ACK generator in C++ (ACK accumulation, duplicate generation, proactive loss masking) to sustain high goodput (70% of link/70 Mbps) under heavy bi-directional burst conditions while accelerating recovery, preventing TCP timeouts, and eliminating tail loss in low-latency networks.
- Conducted extensive ns-3 simulations to optimize TCP flow control and ACK generation mechanics, delivering a fault-tolerant solution that manages TCP flow control, prevents TCP Timeouts, minimizes tail latency, and eliminates excessive redundancy bandwidth waste.

Wireless System Emulation: MU-MIMO Channel Emulator (802.11ac)

Sep. 2015 – Nov. 2016

- **Tools:** Synopsys Symphony, Quartus, Modelsim, MATLAB, Simulink.
- Developed a 4×4 MU-MIMO real-time embedded hardware channel emulator, utilizing MATLAB and Simulink for applied physics modeling of 35 propagation taps and 80 MHz bandwidths.
- Optimized hardware resource allocation to slash Stratix IV FPGA logic utilization from a theoretical 16,800% down to 20%, leveraging Synopsys Symphony to emulate and consolidate parallel filtering architectures onto a single die.

FPGA-Based 2×2 MIMO-OFDM SDM Transceiver

Sep. 2011 – Dec. 2012

- **Tools:** Quartus, Altera DSP Builder, MATLAB, Simulink.
- Architected a complete 2×2 Spatial Division Multiplexing baseband transceiver, driving the design through FPGA prototyping and emulation to achieve a low-latency 400 Mbps datapath.
- Engineered high-performance DSP pipelines (256-point FFT/IFFT, 16-QAM) and FEC architectures (Viterbi decoding), developing custom pipelined 18-bit fixed-point arithmetic units to eliminate fixed-point mathematical overflow.

PUBLICATIONS, GRANTS & AWARDS

- **Publications:** Authored over 40 peer-reviewed international papers (IEEE/IEICE/Scopus), including 7 journal articles, specializing in transport layer optimization, P4 programmable data planes, congestion control, packet loss recovery, and wireless hardware emulation.
- **Research Grants:** Principal Investigator (PI) and Key Lead across 10+ competitive grants funded by Japanese (JSPS KAKEN) and Vietnamese governments/industry, securing over \$100,000+ in project funding.
- **Recognitions & Awards:** Recipient of 3 international conference Best Paper Awards (2019, 2020, 2022), Cisco Instructor Years of Service Award (Mentored 2nd Place Cisco NetRiders National Student Team, 2013), and multiple University Merit Awards.

CERTIFICATIONS

- **Cisco Systems:** CCNA, CCNA Security, CCAI (Cisco Certified Academy Instructor) for CCNA & CCNA Security
- **Microsoft:** MCSE (Systems Engineer) & MCSA (Systems Administrator) – Network Infrastructure & Systems

EDUCATION

Kyushu Institute of Technology

Ph.D., Computer Science and Systems Engineering

Apr. 2014 – Mar. 2017

Fukuoka, Japan

University of Science

M.S. & B.S., Electronics and Telecommunications Engineering

Sep. 2005 – Sep. 2012

Ho Chi Minh City, Vietnam